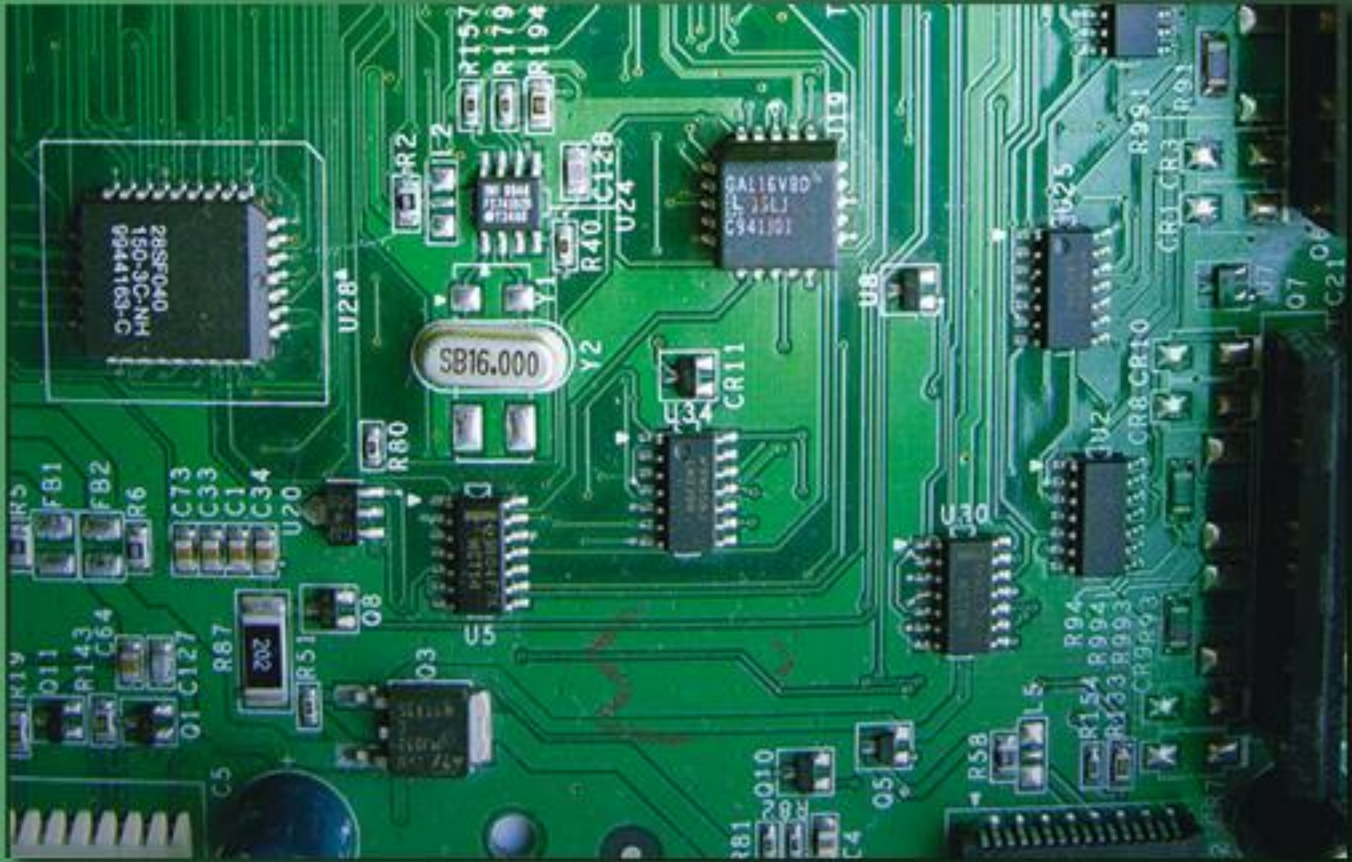
The Intel Microprocessors

8086/8088, 80186/80188, 80286, 80386, 80486 Pentium, Pentium Pro Processor, Pentium II, Pentium 4, and Core2 with 64-bit Extensions

Architecture, Programming, and Interfacing



EIGHTH EDITION

Barry B. Brey

PEARSON

Chapter 12: Interrupts

12–1 BASIC INTERRUPT PROCESSING

- This section discusses the function of an interrupt in a microprocessor-based system.
- Structure and features of interrupts available to Intel microprocessors.

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The Purpose of Interrupts

- Interrupts are useful when interfacing I/O devices at relatively low data transfer rates, such as keyboard inputs, as discussed in Chapter 11.
- Interrupt processing allows the processor to execute other software while the keyboard operator is thinking about what to type next.
- When a key is pressed, the keyboard encoder debounces the switch and puts out one pulse that interrupts the microprocessor.

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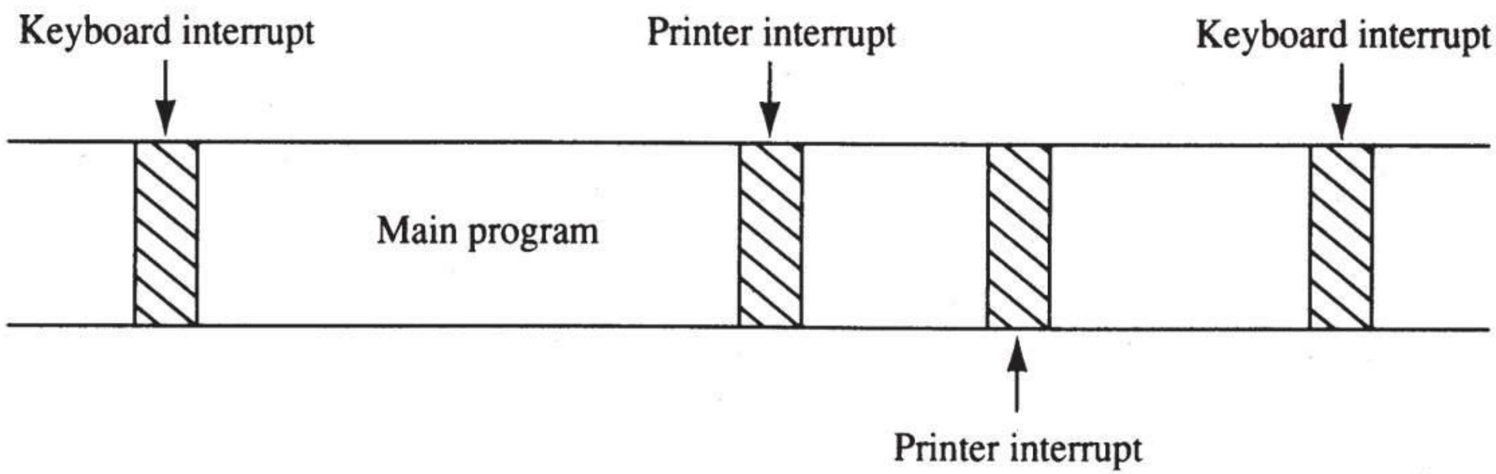
Figure 12–1 A time line that indicates interrupt usage in a typical system.



- a time line shows typing on a keyboard, a printer removing data from memory, and a program executing
- the keyboard interrupt service procedure, called by the keyboard interrupt, and the printer interrupt service procedure each take little time to execute

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Interrupts

- Intel processors include two hardware pins (INTR and NMI) that request interrupts…
- And one hardware pin ($\overline{\text{INTA}}$) to acknowledge the interrupt requested through INTR.
- The processor also has software interrupts INT, INTO, INT 3, and BOUND.
- Flag bits IF (interrupt flag) and TF (trap flag), are also used with the interrupt structure and special return instruction IRET
  - IRETD in the 80386, 80486, or Pentium

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Interrupt Vectors

- Interrupt vectors and the vector table are crucial to an understanding of hardware and software interrupts.
- The **interrupt vector table** is located in the first 1024 bytes of memory at addresses 000000H–0003FFH.
  - contains 256 different four-byte interrupt vectors
- An interrupt vector contains the address (segment and offset) of the interrupt service procedure.

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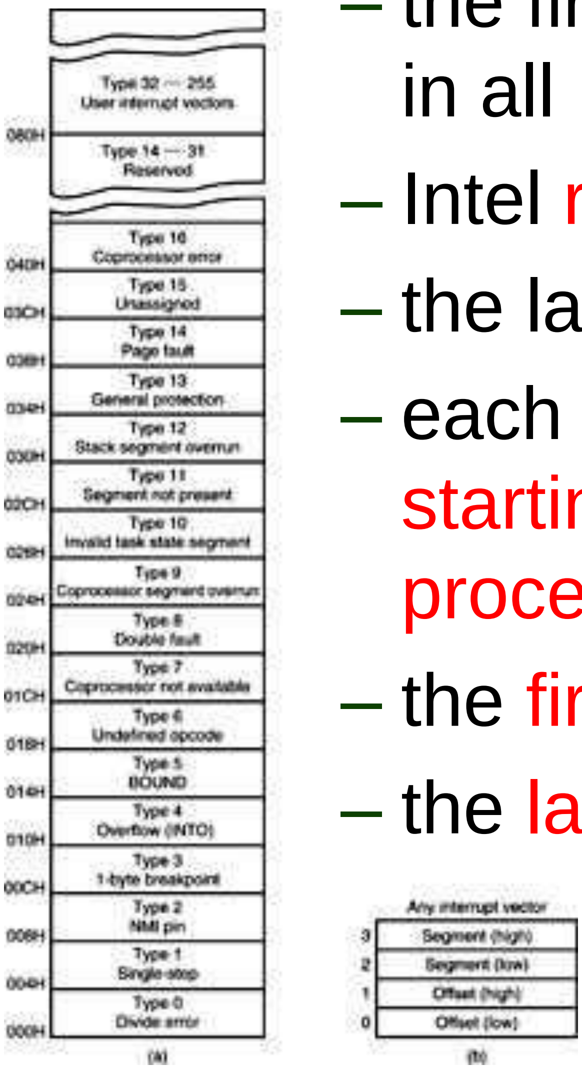
Figure 12–2 (a) The interrupt vector table for the microprocessor and (b) the contents of an interrupt vector.



– the first five interrupt vectors are identical in all Intel processors

– Intel reserves the first 32 interrupt vectors

– the last 224 vectors are user-available

– each is four bytes long and contains the starting address of the interrupt service procedure.

– the first two bytes contain the offset address

– the last two contain the segment address

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Intel Dedicated Interrupts

- **Type 0**
  The **divide error** whenever the result from a division overflows or an attempt is made to divide by zero.
- **Type 1**
  Single-step or trap occurs after execution of each instruction if the trap (TF) flag bit is set.
  - upon accepting this interrupt, TF bit is cleared so the interrupt service procedure executes at full speed

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- **Type 2**
  The **non-maskable interrupt** occurs when a logic 1 is placed on the NMI input pin to the microprocessor.
  - non-maskable—it cannot be disabled
- **Type 3**
  A special one-byte instruction (INT 3) that uses this vector to access its interrupt-service procedure.
  - often used to store a breakpoint in a program for debugging

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- **Type 4**
  Overflow is a special vector used with the INTO instruction. The INTO instruction interrupts the program if an overflow condition exists.
  – as reflected by the overflow flag (OF)

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- **Type 5**
  The **BOUND** instruction compares a register with boundaries stored in the memory.

  If the contents of the register are greater than or equal to the first word in memory and less than or equal to the second word, no interrupt occurs because the contents of the register are within bounds.
  - if the contents of the register are out of bounds, a type 5 interrupt ensues

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Interrupt Instructions: BOUND, INTO, INT, INT 3, and IRET

- Five software interrupt instructions are available to the microprocessor:
- INT and INT 3 are very similar.
- BOUND and INTO are conditional.
- IRET is a special interrupt return instruction.

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- **BOUND** : Check array against boundary
- **INTO: Interrupt on overflow**
- **INT: Interrupt**
- **INT 3: Interrupt 3**
- **IRET: Return from interrupt**

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- BOUND has two operands, and compares a register with two words of memory data.
- INTO checks or tests the overflow flag (O).
  - If O = 1, INTO calls the procedure whose address is stored in interrupt vector type 4
  - If O = 0, INTO performs no operation and the next sequential program instruction executes
- The INT *n* instruction calls the interrupt service procedure at the address represented in vector number *n*.

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- INT 3 instruction is often used as a breakpoint-interrupt because it is easy to insert a one-byte instruction into a program.
  - breakpoints are often used to debug software
- The IRET instruction is a special return instruction used to return for both software and hardware interrupts.
  - much like a far RET, it retrieves the return address from the stack

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12–2 HARDWARE INTERRUPTS

- The two processor hardware interrupt inputs:
  - non-maskable interrupt (NMI)
  - interrupt request (INTR)
- When NMI input is activated, a type 2 interrupt occurs
  - because NMI is internally decoded
- The INTR input must be externally decoded to select a vector.

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- Any interrupt vector can be chosen for the INTR pin, but we usually use an interrupt type number between 20H and FFH.
- Intel has reserved interrupts 00H - 1FH for internal and future expansion.
- $\overline{\text{INTA}}$ is also an interrupt pin on the processor.
  - it is an output used in response to INTR input to apply a vector type number to the data bus connections D_7–D_0
- Figure 12–5 shows the three user interrupt connections on the microprocessor.

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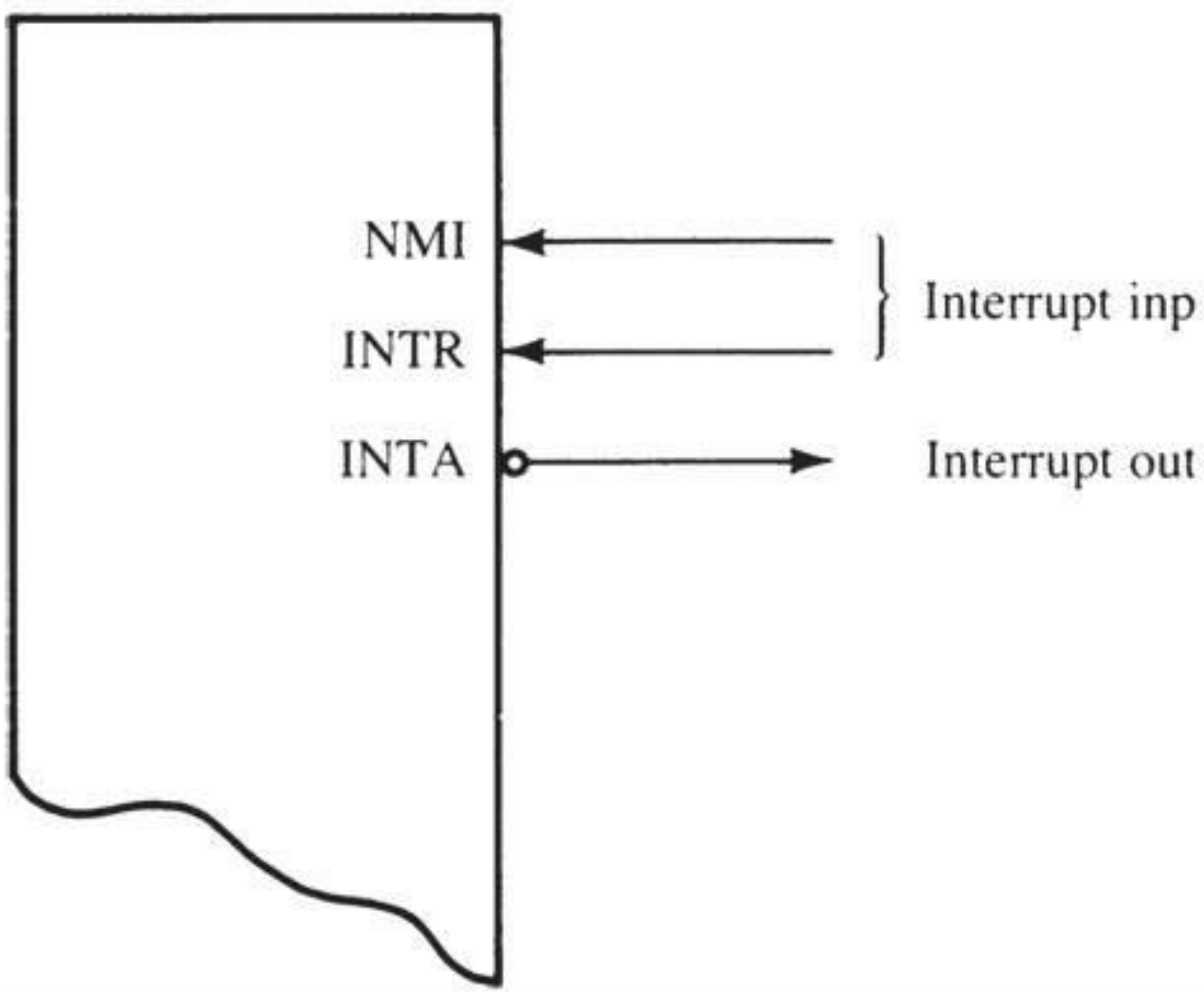
Figure 12–5 The interrupt pins on all versions of the Intel microprocessor.



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INTR and $\overline{\text{INTA}}$

- The interrupt request input (INTR) is level-sensitive, which means that it must be held at a logic 1 level until it is recognized.
  - INTR is set by an external event and cleared inside the interrupt service procedure

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- The processor responds to INTR by pulsing $\overline{\text{INTA}}$ output in anticipation of receiving an interrupt vector type number on data bus connections D_7–D_0.
- Fig 12–8 shows the timing diagram for the INTR and pins of the microprocessor.
- Two $\overline{\text{INTA}}$ pulses generated by the system insert the vector type number on the data bus.
- Fig12–9 shows a circuit to appy interrupt vector type number FFH to the data bus in response to an INTR.

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Figure 12–8 The timing of the INTR input and $\overline{\text{INTA}}$ output. *This portion of the data bus is ignored and usually contains the vector number.

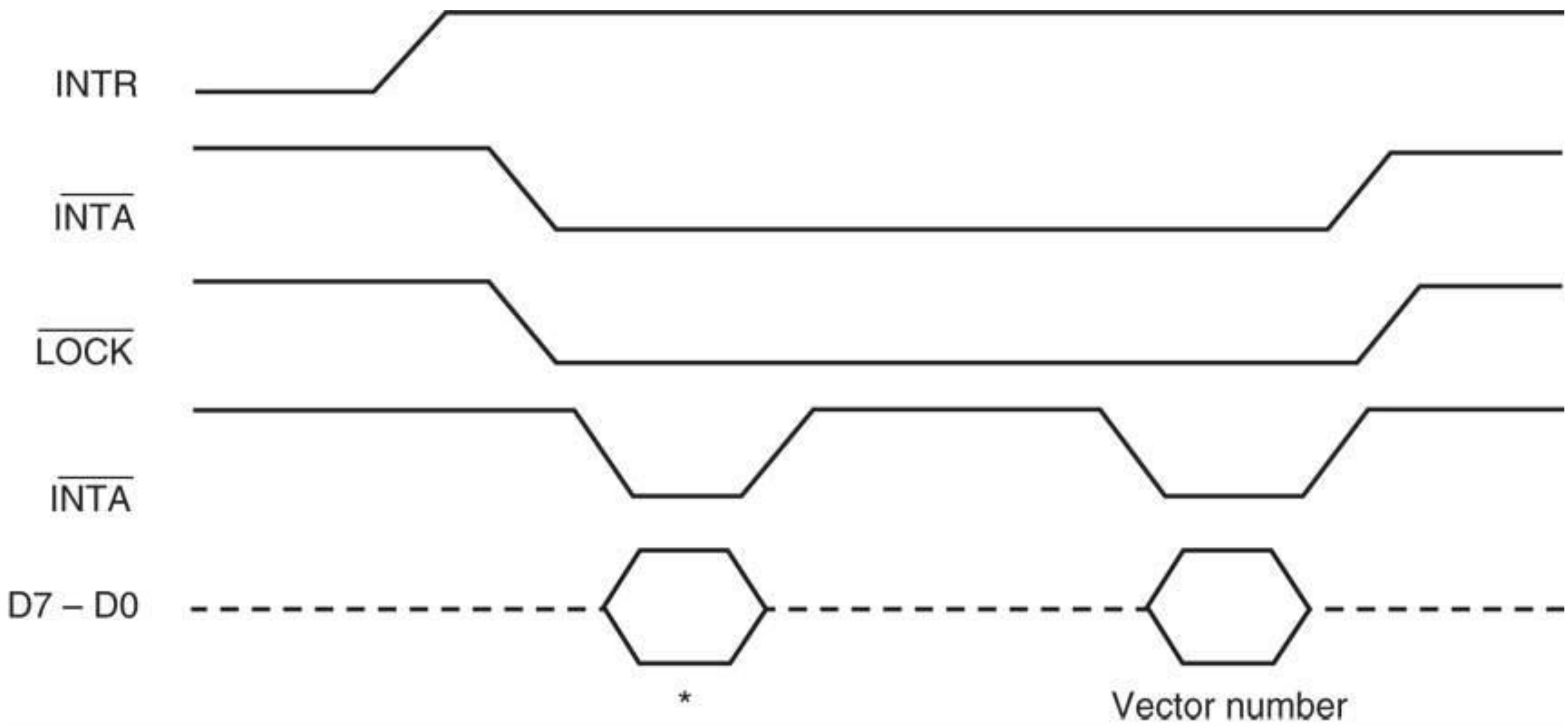

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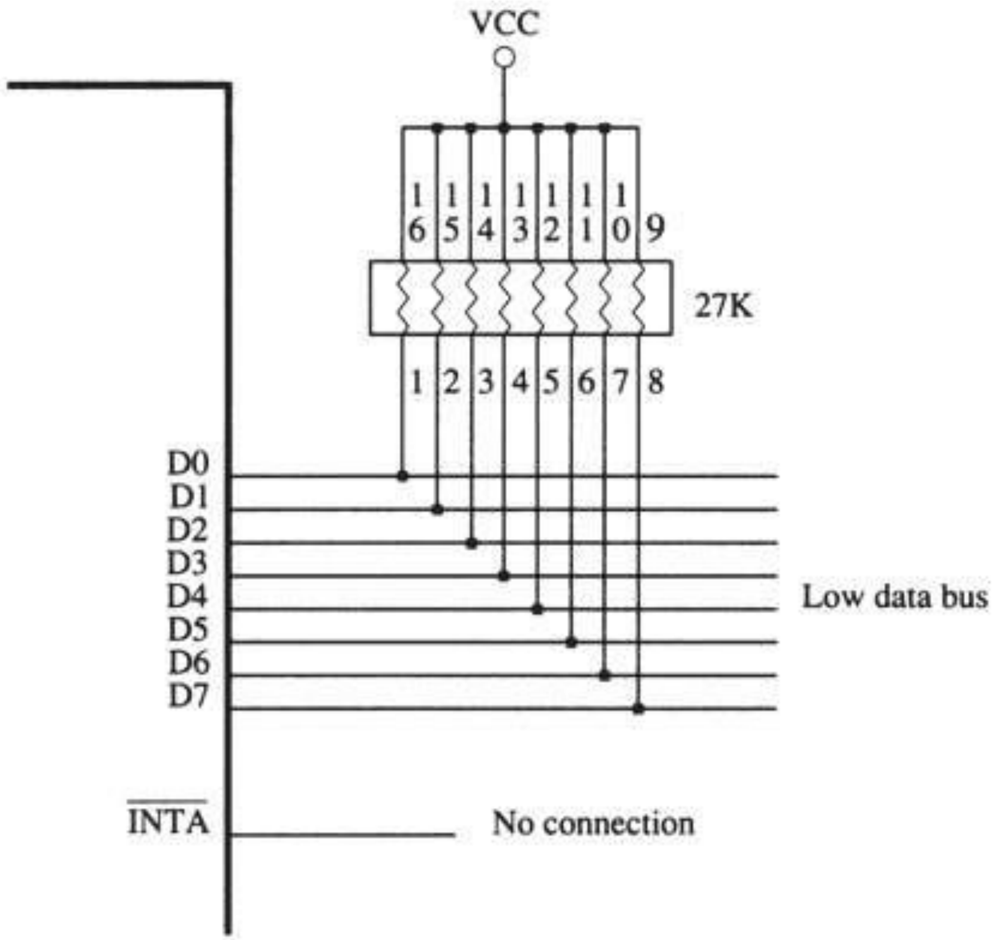
Figure 12–9 A simple method for generating interrupt vector type number FFH in response to INTR.



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Using a Three-State Buffer for INTA

- Fig 12–10 shows how interrupt vector type number 80H is applied to the data bus (D_0–D_7) in response to an INTR.
- In response to INTR, the processor outputs the $\overline{\text{INTA}}$ to enable a 74ALS244 three-state octal buffer.
- The octal buffer applies the interrupt vector type number to the data bus in response.
- The vector type number is easily changed with DIP switches shown in this illustration.

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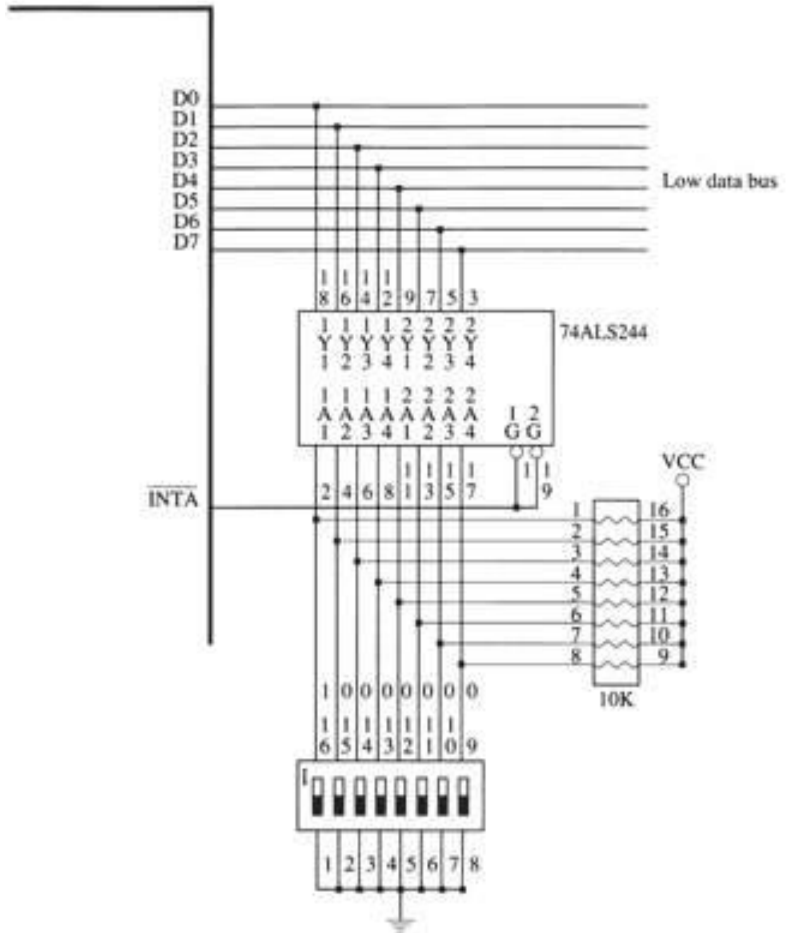
Figure 12–10 A circuit that applies any interrupt vector type number in response to $\overline{\text{INTA}}$. Here the circuit is applying type number 80H.



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